MIL-C-24643/4B
3 October 1986
SUPERSEDING
MIL-C-0024643/4A(SH)
8 November 1985
MIL-C-24643/4
28 September 1984

MILITARY SPECIFICATION SHEET

CORD, ELECTRICAL, 600 VOLTS, TYPE LSMCOS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cord described herein shall consist of this specification and the latest issue of MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, uncoated (see table I for size).
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Ethylene propylene rubber or cross-linked polyethylene insulation (see table I for wall thickness).
- Sizes 2, 4 and 7 shall have special identification code applied by method 3.
- Size 5 shall have one shielded pair (colors black and white) and 3 singles (colors red, green and blue) applied by method 3.
- Size 6 shall have two shielded pairs (colors black and white for first pair and blue and red for second pair) and two singles (colors yellow and green) applied by method 3.
Fourth - Braided shield (for all pairs) of AWG no. 34 or no. 36 uncoated copper. Braid angle 30 to 35 degrees; minimum coverage of 88 percent.
Fifth - The required number of singles and shielded pairs cabled with a left-hand lay and with a minimum amount of fillers to make a round assembly. The length of lay shall be determined to permit the completed cord to withstand the twisting and bending endurance test, and for sizes 5 and 6, the mutual inductance tests. The size 7 shall have a center of glass. A single conductor shall not be used as a central core for the cord.

(B) denotes changes.

AMSC N/A FSC 6145
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REQUIREMENTS: (Continued)

Sixth - In size 5 and 6 only, a binder tape applied helically with overlap. In sizes 2, 4 and 7 only: first, a binder; second, a braided shield of AWG no. 34 or no. 36 uncoated copper strands, having a braid angle of 30-degrees minimum, and providing a coverage of 88-percent minimum; third, a separator tape.

Seventh - Cross-linked polyolefin jacket (black). (See table I for diameters).

NOTE 1: At the manufacturer's option, the conductors in type and size LSMCOS-6 may consist of eight strands, each having a nominal diameter of 0.010 inch, around one strand having a nominal diameter of 0.0142 inch.

TABLE I. Details.

Military part no. M24643/4	Type and size	Number of conductors	Conductor size AWG	Insulation thickness nominal (inch)	Overall diameter		Voltage withstand		Insulation resistance per 1000 feet minimum (megohms)	Conductor resistance per 1000 feet maximum (ohms)
					minimum (inch)	maximum (inch)	Conductor to conductor minimum (volts)	Shield to ground minimum (volts)		
-01UO	LSMCOS-2	2	18(class K)	0.023	0.440	0.460	2,000	1,500	200	6.96
-02UO	LSMCOS-4	4	18(class K)	.023	.490	.510	2,000	1,500	200	6.96
-03UN	LSMCOS-5	5	20(class K)	.013	.375	.390	1,000	500	100	11.28
-04UN	LSMCOS-6	6	20(class K)	.013	.460	.480	1,000	1,500	100	11.27
-05UO	LSMCOS-7	7	18(class K)	.023	.575	.595	2,000	1,500	200	6.96

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet at 25°C, maximum..	(see table I)
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor.....	(see table I)
Shield to ground.....	(see table I)
Insulation resistance - megohms/1000 feet, minimum.....	(see table I)
Conductor and shield continuity.....	No failure

Note: Shield continuity applicable to LSMCOS-2, LSMCOS-4 and LSMCOS-7 only.

Group A:	
Visual and dimensional.....	No failure

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group B:	
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum	700
Elongation - percent, minimum	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cord)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Twisting and bending endurance - (see note 2)	
At 75°C, cycles, minimum	6000
At minus 30 ± 2°C, cycles, minimum	2000

NOTE: 2. Roller diameters for bending endurance test shall be as follows:

Cord type and size	Roller diameter (inches, maximum)
LSMCOS-2	1.4
LSMCOS-4	1.5
LSMCOS-5	1.2
LSMCOS-6	1.4
LSMCOS-7	1.8

Cold bending, cord	
At minus 40 ± 2°C	No failure
Group C:	
Mutual inductance (LSMCOS-5 and LSMCOS-6 only)	
Microhenries, maximum	0.0135
Physicals (aged)	
Insulation	
Ethylene propylene rubber	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged, minimum	75
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	80

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EXAMINATION AND TESTS: (Continued)

	Requirements
Jacket (cord)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Cable filler removability	No failure
Heat distortion - percent of unaged, maximum	30
Shield - conformance to material construction and coverage	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Aging and compatibility (cable)(125 ± 5°C)	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	--
(B) Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

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UNIT ORDERING LENGTHS:
All sizes 2000 feet (nominal)

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
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